

PROBABILITY

- **Experiments:** - It is a process of observation in lab, factory, street, nature. But for us experiments involve randomness, chance effects so, that we cannot predict the result exactly.
- **Random Experiment:** A repeatable experiment the results of which are known in advance but exact result will be known only after the experiment is completed is known as random experiment. Example: Tossing of coins, rolling of dice, drawing of a card etc.
- **Outcomes and Trials:** The result of a random experiment is known as outcomes and conduct of a random experiment is known as trial.
- **Trial:** - It is a single performance of an experiment.
- **Outcomes:** - Result of an experiment is called as outcomes.
- **Sample Space:** Set of all possible outcomes is called as sample space.
- **Events:** - Any subset of sample space is called as event.
- **Impossible Events:** ϕ is a subset of sample space S. Hence this is also an event. This event is called as impossible events.
- **Sure Events:** S itself is a subset of sample space S. Hence this is also an event. This event is called as sure events.
- **Simple Event:** An event having only one sample point is called as simple event.
- **Mixed Event:** An event having more than one sample point is called as mixed event.
- **Union of two set:** $-(A \cup B) = \{x: x \in A \text{ or } x \in B\}$
- **Intersection of two set:** $-(A \cap B) = \{x: x \in A \text{ and } x \in B\}$
- **Complement of a set:** $-A^c = S - A = \{x: x \in S \text{ but } x \notin A\}$
- **Mutually exclusive (Disjoint) Events:** - A set of events is said to be mutually exclusive events if happening of one of them prevents the happening of any one of the remaining events. Or If $(A \cap B) = \phi$ then A and B are called as mutually exclusive or disjoint.
- **Mutually exhaustive Events:** - The list of all possible outcomes is said to be mutually exhaustive events if necessarily one of them must happen. Or If $(A \cap B) = \phi$ then A and B are called as mutually exclusive or disjoint.

- **Equally Likely Events:** A set of events is said to be equally likely if there are no reason to expect one of them in preference to the others.

- **Odds in favour of and odd against an Event:** Let A is any event of an experiment.

$$\text{Odd in favour of A} = \frac{P(A)}{P(A^c)} \quad \text{and} \quad \text{Odd Against of A} = \frac{P(A^c)}{P(A)}$$

$$\text{If } \frac{P(A)}{P(A^c)} = \frac{m}{n}, \text{ then } P(A) = \frac{m}{m+n} \text{ and } P(A^c) = \frac{n}{m+n}$$

- Absolute frequency $f(A)$ of an event in n trials is the no. of times a occurs.

$$\text{Relative frequency} = \frac{f(A)}{n}$$

$$f_{rel}(A) = \frac{f(A)}{n}$$

If A not occurs $f(A) = 0$.

If A always occurs $f(A) = n$.

$$\therefore 0 \leq f_{rel}(A) \leq 1$$

- **Probability:** $P(A) = \frac{|A|}{|S|} = \frac{\text{No. of Elements in A}}{\text{No. of Elements in S}}$

- **Probability of Sure Event:** $P(S) = \frac{|S|}{|S|} = 1$.

- **Probability of Impossible Event:** $P(f) = \frac{|f|}{|S|} = \frac{0}{|S|} = 0$.

- $0 \leq P(A) \leq 1$.

- If $A \subseteq B \Rightarrow P(A) \leq P(B)$.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

- If A & B are mutually exclusive events then $A \cap B = f$. So, $P(A \cup B) = P(A) + P(B)$.

- $P(A^c) = 1 - P(A)$.

- In general if there are $A_1, A_2, A_3, A_4, \dots$ are many mutually exclusive events. Then $P(A_1 \cup A_2 \cup A_3 \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

- If A and B are two events: then

$$P(\text{both A, B occurs}) = P(A \cap B)$$

$$P(\text{at least one of A, B occurs}) = P(A \cup B)$$

$$P(\text{non of A, B occurs}) = P(\bar{A} \cap \bar{B})$$

- $P(\text{exactly one of } A, B \text{ occurs}) = P(A \cup B) - P(A \cap B) = P(A) + P(B) - 2P(A \cap B)$
- If A, B and C are three events then,
 $P(\text{at least one of } A, B, C \text{ occurs}) = P(A \cup B \cup C)$
 $P(\text{at least two of } A, B, C \text{ occurs})$

$$= P(\bar{A} \cap B \cap C) + P(A \cap \bar{B} \cap C) + P(A \cap B \cap \bar{C}) + P(A \cap B \cap C)$$

$$= P(A \cap B) + P(B \cap C) + P(C \cap A) - 2P(A \cap B \cap C)$$

 $P(\text{exactly two of } A, B, C \text{ occurs}) = P(\bar{A} \cap B \cap C) + P(A \cap \bar{B} \cap C) + P(A \cap B \cap \bar{C})$

$$= P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$$

 $P(\text{exactly one of } A, B, C \text{ occurs}) = P(\bar{A} \cap \bar{B} \cap C) + P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C})$

$$= P(A) + P(B) + P(C) - 2P(A \cap B) - 2P(B \cap C) - 2P(C \cap A) + 3P(A \cap B \cap C)$$
 - Conditional Probability:** Probability of event B under the condition A occurs is term as conditional probability of B with given A.
 $P(B/A) = \frac{P(A \cap B)}{P(A)}$ where $P(A) \neq 0$.
 Similarly $P(A/B) = \frac{P(A \cap B)}{P(B)}$ where $P(B) \neq 0$.
 - Multiplication Theorem:** If A and B are two events in S. Then
 $P(A \cap B) = P(A) \times P(B/A)$ where $P(A) \neq 0, P(B) \neq 0$
 $P(A \cap B) = P(B) \times P(A/B)$ where $P(A) \neq 0, P(B) \neq 0$
 - Independent Events:** Two events A and B are said to be independent events if occurrence of event A cannot influence the happening of event B. Two events A and B are said to be independent if $P(A \cap B) = P(A) \times P(B)$.
 - If two events A and B are independent then $\bar{A}$ and $\bar{B}$ are independent.
 - If two events A and B are independent then $\bar{A}$ and B are independent.
 - If two events A and B are independent then A and $\bar{B}$ are independent.
 - A set of events $A_1, A_2, A_3, \dots, A_n$ are said to be pair wise independent if $P(A_i \cap A_j) = P(A_i) \times P(A_j)$ for all $i \neq j$.
 - A set of events $A_1, A_2, A_3, \dots, A_n$ are said to be mutually independent if $P(A_i \cap A_j) = P(A_i) \times P(A_j)$ for all $i \neq j$ and $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n)$.
 - Total Probability Theorem:** Let S be the sample space and Let $A_1, A_2, A_3, \dots, A_n$ be 'n' mutually exclusive and exhaustive events associated with random experiment. If E is any event which occurs with A_1 or A_2 or A_3 or ... or A_n then:

$$P(E) = P(A_1) \times P(E/A_1) + P(A_2) \times P(E/A_2) + P(A_3) \times P(E/A_3) + \dots + P(A_n) \times P(E/A_n)$$

$$P(E) = \sum_{r=1}^n P(A_r) \times P(E/A_r)$$

- **Baye's Theorem:** Let S be the sample space and Let $A_1, A_2, A_3, \dots, A_n$ be 'n' mutually exclusive and exhaustive events associated with random experiment. If E is any event which occurs with A_1 or A_2 or A_3 or ... or A_n then:

$$P(A_k / E) = \frac{P(A_k) \times P(E / A_k)}{\sum_{r=1}^n P(A_r) \times P(E / A_r)} \text{ for } k = 1, 2, 3, \dots, n.$$

- The number of exhaustive cases of tossing 'n' coins simultaneously (or tossing a coin 'n' times), the total outcomes = 2^n .
- The number of exhaustive cases of throwing 'n' dice simultaneously or (throwing a dice 'n' times), the total outcomes = 6^n .

- If a coin tossed 'n' times then the probability of getting 'x' heads and 'n - x' tails = $\frac{{}^nC_x}{2^n}$.

- If there are 'n' children in a family then the probability of exactly 'x' boys = $\frac{{}^nC_x}{2^n}$.

- If a coin is tossed 'n' times, then probability of getting odd number of heads = $\frac{1}{2}$.

- If a coin is tossed 'n' times, then probability of getting odd number of tails = $\frac{1}{2}$.

- If a coin is tossed 'n' times, then probability of getting odd number of heads = $\frac{1}{2}$.

- If a coin is tossed 'n' times, then probability of getting at-least one head = $\frac{2^n - 1}{2^n}$.

- If a coin is tossed 'n' times, then probability of getting at-least one tail = $\frac{2^n - 1}{2^n}$.

- When two fair dice are rolled once, then number of favorable cases to the sum of the digits on the faces given by:

Sum	2	3	4	5	6	7	8	9	10	11	12
No. of Cases	1	2	3	4	5	6	5	4	3	2	1

- When three fair dice are rolled once, then number of favorable cases to the sum of the digits on the faces given by:

Sum	3	4	5	6	7	8	9	10
No. of Cases	2C_2	3C_2	4C_2	5C_2	6C_2	7C_2	${}^8C_2 - 3$	${}^9C_2 - 9$

Sum	11	12	13	14	15	16	17	18
No. of Cases	${}^9C_2 - 9$	${}^8C_2 - 3$	7C_2	6C_2	5C_2	4C_2	3C_2	2C_2

- When a fair dice are rolled once the number of favorable cases to get the sum 'r' is the coefficient of x^r in the multinomial expansion of $(x + x^2 + x^3 + x^4 + x^5 + x^6)^n$.

- **Two person's game:** If 'p' and 'q' are the probabilities of success and failure of a game in which A and B play. If A starts the game then:

$$\text{Probability of A wins} = \frac{p}{1-q^2} = \frac{1}{1+q}.$$

$$\text{Probability of B wins} = \frac{qp}{1-q^2} = \frac{q}{1+q}.$$

$$\text{The ratio of their success} = \frac{1}{q}.$$

- If p_A and q_A are the probability of success and failure of A and p_B and q_B are the probability of success and failure of B. Probability of A wins = $\frac{p_A}{1-q_A q_B}$.

- **Two person's game:** If 'p' and 'q' are the probabilities of success and failure of a game in which A, B and C play. If A starts the game then:

$$\text{Probability of A wins} = \frac{p}{1-q^3} = \frac{1}{1+q+q^2}.$$

$$\text{Probability of B wins} = \frac{pq}{1-q^3} = \frac{q}{1+q+q^2}.$$

$$\text{Probability of C wins} = \frac{pq^2}{1-q^3} = \frac{q^2}{1+q+q^2}.$$

$$\text{The ratio of their success} = 1 : q : q^2.$$

- If $r(1 \leq r \leq 7)$ squares are selected at random from a chess board, then the probability that they lie on a diagonal is $\frac{4({}^7C_r + {}^6C_r + {}^5C_r + \dots + {}^rC_r) + 2({}^8C_r)}{{}^{64}C_r}$.

- Out of 'n' pairs of shoes if $r(r < n)$ shoes be selected at random then:

$$\text{The probability that there is no pair} = \frac{{}^nC_r \times 2^r}{{}^{2n}C_r}.$$

$$\text{The probability that there is at-least one pair} = 1 - \frac{{}^nC_r \times 2^r}{{}^{2n}C_r}.$$

$$\text{The probability that there are exactly 'k' pairs} = \frac{{}^nC_k \times {}^{n-k}C_{r-2k} \times 2^{r-2k}}{{}^{2n}C_r}.$$

- **Derangement Problems:** If 'n' letters corresponding to 'n' envelopes with address on them are placed in the envelopes one in each, then:

$$\text{Probability that all the letters are placed in the right envelope} = \frac{1}{n!}.$$

$$\text{Probability that all the letters are not in the right envelope} = 1 - \frac{1}{n!}.$$

$$\text{Probability that all the letters are in the wrong envelope} = \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right).$$

Probability that exactly 'r' letters are in the right envelope

$$= \frac{1}{r!} \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^{n-r} \frac{1}{(n-r)!} \right).$$

- If A and B are two finite sets with number of elements m and n respectively and if a mapping is selected at random from the set of mapping from A to B, then:

Probability that a mapping is one-one (injective) function = $\frac{{}^nP_m}{n^m}$.

Probability that a mapping is many-one function = $1 - \frac{{}^nP_m}{n^m}$.

Probability that a mapping is constant function = $\frac{n}{n^m}$.

Probability that a mapping is bijective function = $\frac{n!}{n^n}$. If (m = n)

Probability of onto function where $m \geq n$

$$= \frac{{}^nC_0 n^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m - {}^nC_3 (n-3)^m + \dots + (-1)^{n-1} {}^nC_{n-1} (1)^m}{n^m}$$

- Out of $2n+1$ tickets consecutively numbered three are selected at random. Then the probability that they are in A.P. is $\frac{3n}{4n^2-1}$.
- If 'n' men among whom A and B stand in a row (line), then the probability that there will be exactly 'r' persons between A and B = $\frac{2(n-r-1)}{n(n-1)}$.
- If 'n' men among whom A and B sit along a circle, then the probability that there will be exactly 'r' persons between A and B = $\frac{2}{n-1}$. Where $r \neq \frac{n-2}{2}$.
- If 'n' men among whom A and B sit along a circle, then the probability that there will be exactly 'r' persons between A and B = $\frac{1}{n-1}$. Where $r = \frac{n-2}{2}$.
- If 'n' men among whom A and B sit along a circle, then the probability that A and B sit together = $\frac{2}{n-1}$.
- If 'n' men among whom A and B sit along a circle, then the probability that A and B never sit together = $1 - \frac{2}{n-1} = \frac{n-3}{n-1}$.
- If 'n' men among whom A and B sit along a circle, then the odd against A and B sit together = $\frac{n-3}{2}$.
- A man has bunch of 'n' keys of which only one fits the door. He tries them successively without repetition. Then, the probability that the door is opened at any trial (less than 'n') is $\frac{1}{n}$.

- If 'n' whole numbers are taken at random and multiplied together then the probability that the last digit of the product is:

$$\text{Either 1, 3, 7, or 9 is} = \frac{4^n}{10^n} = \left(\frac{2}{5}\right)^n.$$

$$\text{Either 2, 4, 6, or 8 is} = \frac{8^n - 4^n}{10^n} = \frac{4^n - 2^n}{5^n}.$$

$$\text{Only 5 is} = \frac{5^n - 4^n}{10^n}$$

$$\text{Only 0 is} = \frac{10^n - 8^n - 5^n + 4^n}{10^n}$$

- E and F are mutually exclusive events and the random experiment is repeated till E or F occurs. Then probability that E occurs before F is $= \frac{P(E)}{P(E) + P(F)}.$
- If A has 'm' shares in a lottery where there are 'm' prizes and 'n' (n > m) blanks then probability of A's winning is $1 - \frac{{}^nC_m}{{}^{m+n}C_m}.$
- Using the vertices of a polygon having 'n' sides a triangle is constructed at random. The probability that the triangle so formed is such that no two sides of the polygon is side of the triangle is $= \frac{(n-4)(n-5)}{(n-1)(n-2)}.$
- Out of 'n' persons sitting at a round table, three persons are selected at random then the probability that no two of them are consecutive is $= \frac{(n-4)(n-5)}{(n-1)(n-2)}.$

RANDOM VARIABLE AND DISTRIBUTION

- **Random Variable:** Let S be a sample space of a random experiment. A real valued function $X : S \rightarrow R$ is called a random variable.
- **Discrete Random Variable:** A random variable which can take only finite number of values is called a discrete random variable.
- **Continuous Random Variable:** A random variable which can take any value between two given limits is called a continuous random variable.
- **Probability Distribution:** Let S be a sample space and $X : S \rightarrow R$ be a random variable. The function $F : R \rightarrow R$ denoted by $F(x) = P(X \leq x)$, is called probability distribution function of the random variable X.
- A set E is said to be countable, if there exist a one – one correspondence between E and the set of natural number N.
- If a sample space is countable then it is called as discrete sample space. A real valued function defined on discrete sample space is called a discrete random variable.

- If $X : S \rightarrow R$ is a discrete random variable with range $\{x_1, x_2, x_3, \dots\}$ then $\sum_{r=1}^n P(X = x_r) = 1$.
- **Mean and Variance of Random Variable:** Let $X : S \rightarrow R$ is a discrete random variable with range $\{x_1, x_2, x_3, \dots\}$ and if $\sum_{r=1}^n x_r P(X = x_r)$ exists then $\sum_{r=1}^n x_r P(X = x_r)$ is called as **mean of the random variable X** . It is denoted by $\bar{x} = E(x)$. If $\sum_{r=1}^n (x_r - \bar{x})^2 P(X = x_r)$ exists, then $\sum_{r=1}^n (x_r - \bar{x})^2 P(X = x_r)$ it is called as **variance of the random variable X** . It is denoted by s^2 . The positive square root of variance is called as standard deviation of the random variable X . It is denoted by s .
Mean = $\bar{x} = E(x) = \sum_{r=1}^n x_r P(X = x_r)$
Variance = $s^2 = \sum_{r=1}^n (x_r - \bar{x})^2 P(X = x_r)$
Standard Deviation = $\sqrt{\text{Variance}}$
- If the range of discrete random variable X is $\{x_1, x_2, x_3, \dots\}$ and $P(X = x_n) = P_n$ for every positive integer n is given then $s^2 + \bar{x}^2 = \sum_{r=1}^n x_r^2 P_r = E(x^2)$.
- **Binomial or Bernoulli's Distribution:** Let n be a positive integer and p be a real number such that $0 \leq p \leq 1$. A random variable X with range $\{0, 1, 2, \dots, n\}$ is said to follow (or have) **Binomial distribution or Bernoulli distribution** with parameters n, p if $P(x = r) = {}^n C_r p^r q^{n-r}$ for $r = 0, 1, 2, 3, \dots, n$ where $p + q = 1$.
- **Mean & Variance of Binomial or Bernoulli's Distribution:** If the random variable X follows a **binomial distribution** with parameters n, p then mean of X is $\text{mean} = np$ and the variance is $\text{variance} = npq$ where $p + q = 1$.
- **Multinomial Expansion:** If a die has m faces marked with $1, 2, 3, \dots, m$ and if such n dice are thrown, then the probability that the sum of the number exhibited on the upper faces equals to p is given by the coefficient of x^p in the expansion of $\frac{(x + x^2 + \dots + x^m)^n}{m^n}$.
- **Poisson's Distribution:** Let $\lambda > 0$ be a real number and mean of the distribution. A random variable X with range $\{0, 1, 2, \dots, n\}$ is said to follow **Poisson distribution** with parameter λ if $P(x = r) = \frac{e^{-\lambda} \lambda^r}{r!}$ for $r = 0, 1, 2, 3, \dots, n$.
- **Mean & Variance of Poisson's Distribution:** If a random variable X follows **Poisson distribution** with parameter λ then $\text{mean} = \text{variance} = \lambda$.
- **Normal Distribution:** The Normal probability distribution is given by $P(X = x) = \frac{1}{s\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\bar{x}}{s}\right)^2}$ where $z = \frac{x-\bar{x}}{s}$ is known as standard variate for $-\infty < x < \infty$.

• **Properties of Normal Distribution:**

- (a) Normal distribution is the limiting case of Binomial Distribution i.e. $B(n, p)$ when $n \rightarrow \infty$.
- (b) For Normal Distribution Mean = Median = Mode.
- (c) Total Area under a standard normal distribution is 1.
- (d) Normal curve is bell-shaped and uni-modal.
- (e) $Q_3 - \text{Median} = \text{Median} - Q_1$.
- (f) $\int_{-1}^1 P(X = r) dx = 1$.
- (g) $P(z_1 \leq z \leq z_2) = P(z_1 < z \leq z_2) = P(z_1 \leq z < z_2) = P(z_1 < z < z_2)$.
